NEUROCOBAL TABLETS 500MCG

NAME AND STRENGTH OF ACTIVE SUBSTANCE:
Each tablet contains:
Methylcobalamin…………………………500mcg.

PRODUCT DESCRIPTION:
A mottling, maroon round of diameter 6mm flat tablet with “MPI” marking.

PHARMACODYNAMICS
(1) Methylcobalamin is a kind of endogenous coenzyme B12:
As a coenzyme of methionine synthetase, Methylcobalamin plays an important role in transmethylation in the synthesis of methionine from homocysteine.

(2) Methylcobalamin is well transported to nerve cell organelles, and promotes nucleic acid and protien synthesis:
Experiments in rats show that Methylcobalamin is better transported to nerve cell organelles than cyanocobalamin and promotes nucleic acid and protein synthesis more than cobamamide does. Experiments with cells from the brain origin and spinal nerve cells in rats also show Methylcobalamin to be involved in the synthesis of thymidine from deoxyuridine, promotion of deposited folic acid utilization and metabolism of nucleic acid.

(3) Methylcobalamin promotes axonal transport and axonal regeneration :
In rat models with streptozotocin-induced diabetes mellitus, Methylcobalamin normalizes axonal skeletal protein transport in sciatic nerve cells. Methylcobalamin exhibits neuropathologically and electrophysiologically inhibitory effects on nerve degeneration in neuropathies induced by drugs, such as adriamycin, acrylamide, and vincristine (in rats and rabbits), models of axonal degeneration in mice and neuropathies in rats with spontaneous diabetes mellitus.

(4) Methylcobalamin promotes myelination (phospholipid synthesis)
Methylcobalamin promotes the synthesis of lecithin which is the main constituent of medullary sheath lipid.
It also increases myelination of neurons in rat tissue culture more than cobamamide does.

(5) Methylcobalamin restores delayed synaptic transmission and diminished neurotransmitters back to normal:
Methylcobalamin restores end-plate potential induction early by increasing nerve fiber excitability in the crushed sciatic nerve in rats. In addition, Methylcobalamin normalizes diminished levels of acetylcholine in brain tissue of rats fed with a choline-deficient diet.

PHARMACOKINETICS:
(1) Absorption
Vitamin B12 substances bind to intrinsic factor, a glycoprotein secreted by the gastric mucosa, and are then actively absorbed from the gastrointestinal tract. Absorption is impaired in patients with an absence of intrinsic factor, with a malabsorption syndrome or with disease or abnormality of the gut, or after gastrectomy. Absorption from the gastrointestinal tract can also occur by passive diffusion; little of the vitamin present in food is absorbed in this manner although the process becomes increasingly important with larger amounts such as those used therapeutically.
(i) Single-dose administration
When Methylcobalamin was administered orally to healthy adult male volunteers at single doses of 120 μg and 1,500 μg) during fasting, the peak serum total vitamin B12 concentration was reached after 3hrs for both doses, and this was dose-dependent.
Note) A single dose of 1,500 μg is unapproved.
(ii) Repeated-dose administration
When Methylcobalamin was administered orally to healthy adult male volunteers at a dose of 1,500 μg daily for 12 consecutive weeks, the serum concentration increased for the first 4 weeks after administration, rising to about twice as high as the initial value. Thereafter, there was a gradual increase which peaked at about 2.8 times the initial value at the 12th week of dosing. The serum concentration declined after the last administration (12 weeks), but was still about 1.8 times the initial value 4 weeks after the last administration.

(2) Distribution
Vitamin B12 is extensively bound to specific plasma proteins called transcobalamins; transcobalamin II appears to be involved in the rapid transport of the cobalamins to tissues. Vitamin B12 is stored in the liver. Vitamin B12 diffuses across the placenta and also appears in breast milk.

(3) Excretion
Vitamin B12 is excreted in the bile, and undergoes extensive enterohepatic recycling; part of a dose is excreted in the urine, most of it in the first 8 hours: urinary excretion, however, accounts for only a small fraction in the reduction of total body stores acquired by dietary means.

40 – 80% of the cumulative amount of total B12 excreted in the urine by 24 hrs after single-dose administration was excreted within the first 8 hrs.

(4) Elimination Half-life
12.5 hrs (single-dose oral administration: calculated from the average of 24-48 hours values)

INDICATION:
Peripheral neuropathies.

RECOMMENDED DOSAGE:
Adult Dosage
The usual adult dosage for oral use is 3 tablets. (1,500µg of Methylcobalamin) daily divided into three doses. The dosage may be adjusted depending on the patient's age and symptoms.

ROUTE OF ADMINISTRATION:
Oral.

CONTRAINDICATIONS:
Hypersensitivity to Methylcobalamin or others components of the formulation.

WARNINGS AND PRECAUTIONS:

This products should not be used aimlessly for more than one month unless it is effective.

Vitamin B12 should, if possible, not be given to patients with suspected vitamin B12 deficiency without first confirming the diagnosis. Where it is desirable to start therapy immediately, combined treatment for both deficiencies may be started once suitable samples have been taken to permit diagnosis of the deficiency and the patient converted to the appropriate treatment once the cause of the anaemia is known. Regular monitoring of the blood is advisable.

Although the haematological symptoms of B12 deficiency and folate deficiency are similar, it is important to distinguish between them since the use of folate alone in B12-deficient megaloblastic anaemia can improve haematological symptoms without preventing aggravation of accompanying neurological symptoms, and may lead to severe nervous system sequelae such as subacute combined degeneration of the spinal cord. Use of doses greater than 10 mictograms daily may produce a haematological response in patients with folate deficiency and indiscriminate use may mask the precise diagnosis. Conversely, folate may mask vitamin B12 deficiency.

Precautions concerning use

(1) Administration : Methylcobalamin is susceptible to photolysis. Light decreases the content of Methylcobalamin and tablets may change colour (eg. turn reddish) with exposure to moisture.
Therefore, this product should be used promptly after the package is opened, and caution should be taken so as not to expose the tablets/capsules to light/moisture.

(2) Caution in handling over drug : For drugs that are dispensed in a press-through package (PTP), instruct the patient to remove the drug from the package prior to use.[It has been reported that, if the PTP sheet is swallowed, the sharp corners of the sheet may puncture the esophageal mucosa, causing perforation and resulting in serious complications such as mediastinitis.]

Other Precautions
The prolonged use of larger doses of Methylcobalamin is not recommended for patients whose occupation requires the handling of mercury or mercury compounds.

INTERACTIONS WITH OTHER MEDICAMENTS:
Absorption of vitamin B12 from the gastrointestinal tract may be reduced by neomycin, aminosalicylic acid, histamine H2-antagonists, omeprazole, and colchicine.
Serum concentrations may be decreased by use of oral contraceptives.
Many of these interactions are unlikely to be of clinical significance but should be taken into account when performing assays for blood concentrations.

PREGNANCY AND LACTATION:
Pregnancy
There are no data available for Methylcobalamin to be used in pregnant women.
Lactation
There are no data available for Methylcobalamin to be used in lactating women. However, since vitamin B12 is distributed into breast milk, The American Academy of Pediatrics considers its use to be usually compatible with breast feeding.

SIDE EFFECTS:

Dermatologic Effects: Rash ; In the event of such symptoms, treatment should be discontinued.
Gastrointestinal Effects: Anorexia, nausea/vomiting and diarrhea.
Neurologic Effects (Central nervous system) : Headache
Others:

- Anaphylactoid reaction : decrease in blood pressure or dyspnea, may occur. Patients should be carefully observed. In the event of such symptoms, treatment should be discontinued immediately and appropriate measures taken.
- Hot sensation
- Diaphoresis
- Pain/induration at the side of intramuscular injection

SYMPTOMS AND TREATMENT OF OVERDOSE:

There have been no reports, in the literature, of overdosage with Methylcobalamin.

EFFECT ON ABILITY TO DRIVE AND USE MACHINE:

Not applicable.

PRECLINICAL SAFETY DATA:

Not applicable.

INSTRUCTION FOR USE:

For oral use only.

STORAGE CONDITIONS:

Store in the original package and in a dry place. Do not store above 30ºC. Protect from light.
Keep out of reach of children/ jauhi dari kanak-kanak.

DOSAGE FORMS AND PACKAGING AVAILABLE:

Dosage form: Tablet
Packing size: 50 x 10’s (blister pack).

NAME AND ADDRESS OF MANUFACTURER / PRODUCT REGISTRATION HOLDER:

MALAYSIAN PHARMACEUTICAL INDUSTRIES SDN BHD (101323-U)
Plot 14, Lebuhraya Kampung Jawa,11900 Bayan Lepas,
Pulau Pinang, Malaysia.

DATE OF REVISION:

12/12/2017